higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MASONRY
NQF LEVEL 2

(12020152)

22 November 2019 (Y-Paper)
13:00–16:00

This question paper consists of 9 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on NEW page.
5. Use only BLUE or BLACK ink.
6. Write neatly and legibly.

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QUESTION 1: GENERAL

1.1 Give ONE word or term for each of the following descriptions by choosing a word or term from the list below. Write only the word or term next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

concrete; mortar; toothing; racking back; an anchor; structural wall

1.1.1 Part of the superstructure of a building and it is a load-bearing wall

1.1.2 A method that creates steps in a wall that will be used to join new brickwork

1.1.3 Used for tying metal straps to masonry or concrete

1.1.4 A mixture of fine aggregates and coarse aggregates

1.1.5 A mixture of fine aggregates and cement

(5 × 1) (5)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write 'True' or 'False' next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

1.2.1 DPC is a barrier in a wall that prevents moisture through the wall.

1.2.2 The thickness of a half-brick wall is 140 mm.

1.2.3 The space in a cavity wall cannot be more than 50 mm.

1.2.4 The floor plan shows the contractor how thick the walls of a specific building will be.

1.2.5 Lintels must be propped for a period of not less than seven days after completion of installation.

(5 × 1) (5)

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1.3 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.3.1–1.3.5) in the ANSWER BOOK.

1.3.1 Which ONE of the following tools can be used when plastering a wall?

A Steel square
B Gauge rod
C Bricklayer's trowel
D Steel pegs

1.3.2 The variance of mortar joints are between …

A 10 mm and 22 mm thick
B 8 mm and 12 mm thick
C 12 mm and 20 mm thick
D 8 mm and 20 mm thick

1.3.3 Concrete lintels should be installed with a minimum of … end bearing at both ends.

A 50 mm
B 100 mm
C 150 mm
D 300 mm

1.3.4 A groove that is inserted into a concrete surface to control cracking.

A Bed joint
B Control joint
C Perpend joint
D Horizontal joint

1.3.5 The bond that usually has three courses of stretchers to one course of headers.

A Stretcher bond
B Header bond
C Garden wall bond
D English bond

(5 × 1) (5)
[15]

QUESTION 2: INTRODUCTION AND BACKGROUND

2.1 Masonry units consist of bricks and blocks that are used in construction.

What is the difference between a *clay brick* and a *concrete block*? (4)

2.2 Define the terminology *masonry unit*. (1)

2.3 What is the height of a standard brick? (1)
[6]

QUESTION 3: SETTING OUT MASONRY WALLS

3.1 State any FOUR ways to check if the corners of the internal and external walls are square. (4)

3.2 In order to do any job correctly, it is very important to choose the correct tools for that specific job. Below is a list of tools that can be used on a construction project.

Bricklayer's trowel
Hand hawk
Builder's square
Spirit level
Racking tool
Steel pegs
Straight edge
Block brush
Bolster
Four-pound hammer
Inside-corner tool
Joint ruler
Plastering trowel

3.2.1 List FOUR tools from the list above that are used for plastering a blank wall. (4)

3.2.2 Give FOUR tools that can be used for setting out a building. (4)

3.2.3 Name TWO tools that can be used for finishing brickwork. (2)

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3.3 The following is the floor plan of a dwelling which is to be constructed. Study the plan and answer the question.



3.3.1 Bedroom 1 and bedroom 2 are divided by an internal half-brick wall according the drawing.

If the contractor did not build the dividing wall as per plan, would it weaken the structure of the house in terms of the compressive force? Explain your answer. (4)

3.3.2 Give the length and width of the double garage as per the floor plan. (2)

3.3.3 Explain why you could not begin to set up your profiles using only this floor plan. (2)

[22]

QUESTION 4: BASIC CONSTRUCTION OF MASONRY WALLS

4.1 Explain how DPC should be installed or laid in the following walls:

4.1.1 Cavity walls

4.1.2 Free-standing garden walls

4.1.3 Block building walls

(3 × 2) (6)

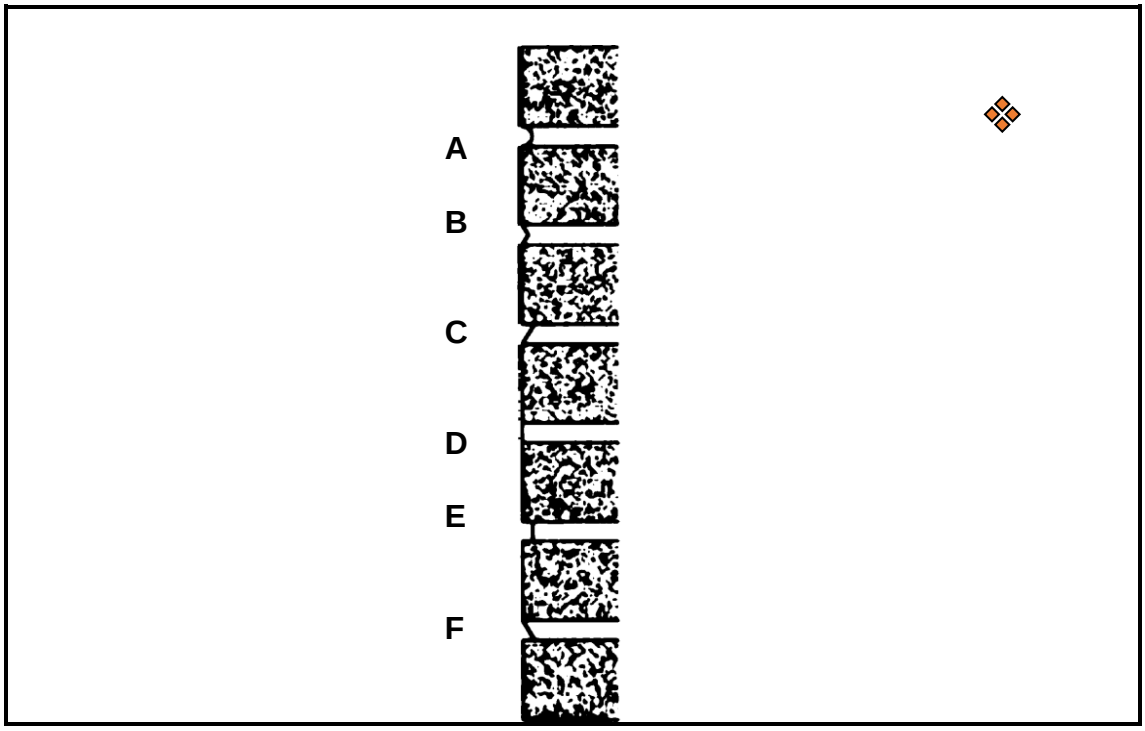
4.2 Why do we use DPC in structural walls? (2)

4.3 Explain the difference between a *single-leaf* wall and a *cavity* wall. (4)

4.4 State another name for the term *brick force*. (1)

4.5 Explain the importance of using brick force at intervals during the construction of a house. (4)

4.6 The construction of face brick walls requires joint profiles which are created by using a tool called a jointer. The diagram below shows the different ways in which mortar joints can be finished.



Give the correct type of mortar finishes for (A–F) next to the question number (4.6) in the ANSWER BOOK. (6)

4.7 List THREE solid unit sizes that are appropriate for stretcher, English, Flemish and garden wall bonds. (3)

4.8 What is a *weep hole*? (1)

4.9 Which wall ties are preferred in concrete block walls? (1)

[28]

QUESTION 5: ADVANCED CONSTRUCTION OF MASONRY WALLS

5.1 Choose a/an item/word from COLUMN B that matches a description in COLUMN A. Write only the letter (A–L) next to the question number (5.1.1–5.1.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
5.1.1	The steel in the concrete was put under tensile strain when units were made	A	structural wall
		B	roof anchor
5.1.2	Damp-proof course	C	lintel
5.1.3	A load-bearing wall	D	toothing
5.1.4	A piece of timber, stone or concrete over a door, window or opening in a building	E	non-structural wall
		F	DPM
5.1.5	Ties a wall to the roof trusses	G	pre-stressed concrete
5.1.6	Controls movement in masonry	H	expansion joints
5.1.7	The exclusion of half brick every alternate course	I	DPC
		J	weep holes
5.1.8	A long piece consisting of two or more materials joined together	K	composite beam
5.1.9	Designed to drain away moisture	L	centering
5.1.10	The temporary supporting of arches		

(5 × 2) (10)

5.2 Explain the procedures that will be followed when installing a pre-stressed concrete lintel. (8)

5.3 List THREE factors that are considered when specifying roof ties. (3)

5.4 Give TWO functions of wall ties in a double-leaf wall. (2)

5.5 List THREE main steps to lay down a damp-proof course under a window sill exposed to the weather in a solid masonry single-leaf wall. (3)

5.6 State THREE places were control joints can be formed in brickwork. (3)

[29]

TOTAL 100

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